16 Apr '26

OS

Memory Management

→ Non - Continuous Memory — Paging

8 bit
1 bit

1 Byte = 8 bits

Word
↳ 8 bits
↳ 16 bits
↳ 32 bits (common)

$1\,KB \rightarrow 2^{10}$ bytes
$1\,MB \rightarrow 2^{20}$ bytes
$1\,GB \rightarrow 2^{30}$ bytes
$1\,TB \rightarrow 2^{40}$ bytes
$1\,PB \rightarrow 2^{50}$ bytes
~~1 X~~

Q 16 bit address → size of location 1 bytes

↳ $2^{10} \times 1$ bytes
1KB

Q 44 bit address → size of location 2 bytes

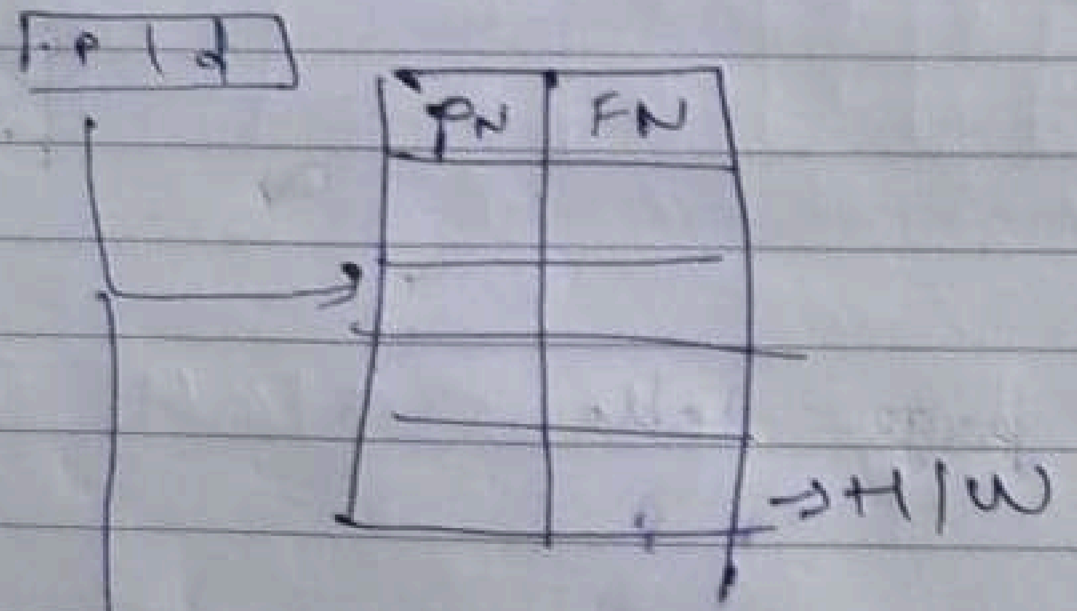
↳ $2^{44} \times 2$ bytes
$2^{4}\,T \times 2$ bytes
32TB

Q 64 GB memory
↳ 32 bit address

$\cancel{2^{32}}\ 16\ \dfrac{64\,GB \times \cancel{2^{30}}}{\cancel{4}\cdot\cancel{2^{32}}} = 16$ bytes

Locality of Reference
↳ Spatial Locality - Space
↳ Temporal Location - time

Translation Lookaside Buffer → stores frame, page number so can access directly.



→ Page Table (If not found in TLB)

$$\text{EMAT} = \overset{\text{Hit Ratio}}{H}(\text{TLB} + \text{MMAT}) + (1 - H)(\text{TLB} + 2\text{MMAT})$$
(with TLB).

$$\text{EMA}_{T} = 2\text{MMAT}$$

Q
MMAT = 400 micro seconds
TLBT = 50 micro seconds
Hit ratio = 90%.

$$\text{EMAT} = \cancel{\tfrac{90}{100}}\,0.9\,(50 + 400) + \cancel{b}\,0.1\,(50 + 800)$$
$$= 0.9 \times 450 + 0.1 \times 850$$
$$= 405 + 85$$
$$= 490$$

$$\text{EMAT} = 800$$

Ans

$$\frac{6}{10} \times \frac{5}{9} + \frac{4}{10} \times \frac{3}{9}$$

$$\Rightarrow \frac{30+12}{90} \Rightarrow \frac{42}{90} = \frac{7}{15}$$

~~Ans~~

21st Apr '26

OS

Paging / Non continous memory allocation

Main Memory size → Physical Address space } Frames

Logical Address space → Secondary Memory size } Pages

p	d

↓ Page No. └ Page ~~offset~~/size

f	d

↓ Frame No └ Frame offset / Size

Q

No. of pages $= \cancel{32}\ 2^{20}$

Page size $= 2B$

Size Memory ↳ $2^{20} \times 2B$

2 MB

Q SM size = 64 GB

Word addressable memory = 4B

No. of pages $= \frac{64\,GB}{4B} \rightarrow \frac{\cancel{64}^{16} \times 2^{30} B}{4B}$

$= 16 \times 2^{30} = 2^{34}$

Q

L.A = 32 bit
P.A = 16 bit
No. of frames = 4 bit
No. of pages = ~~30 bit~~ ~~18 bit~~ 20 bit

P.A

4	2^{12}

16

LA Frame size = 2^{12}

20	12

32

22nd Apr'26

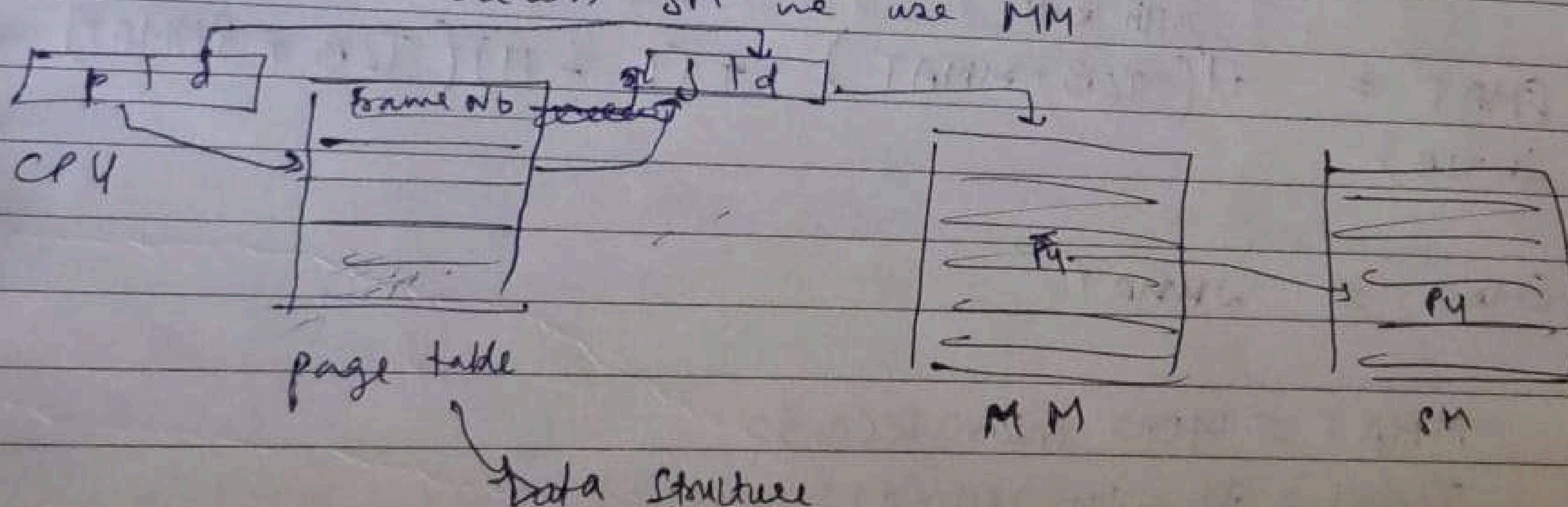
OS

Address Mapping

Every process has it's own page table

Address Translation

↳ To access SM we use MM



Effective Memory Access Time = 2× Main Memory Access Time

Size of Page Table = Number of process
= Page Table Entry Size

Main memory size = No. of frames × frame size

[Page size = Frame size]

Q. MM = 32 MB

No. of Frame = 2048

$$\text{Frame size} = \frac{32 \times 2^{20}\ B}{2048 = 2^{11}} = 2^{14}\ B = 16\ KB$$

Q. MM = 256 MB

SM = 512 MB

No. of Frames = 4096

$$\text{Frame size} = \frac{\overset{2^8}{256} \times 2^{20}}{2^{12}} = \underbrace{2^{16}\ B} = 32\ KB$$

Page size = 32 KB

$$\text{No. of Pages} = \frac{2^9 \times 2^{30}}{\cancel{2^{16}}\ 2^{14}} = 2^{23}$$

Q. Logical ~~Physical~~ A = 64 Bit

No. of Pages = 64

Page size = 2^{58}

L.A

6 p	d 58

64